



1. Description

1.1. Project

Project Name	EIC_TuckRun_Hall_v6_4
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	02/02/2026

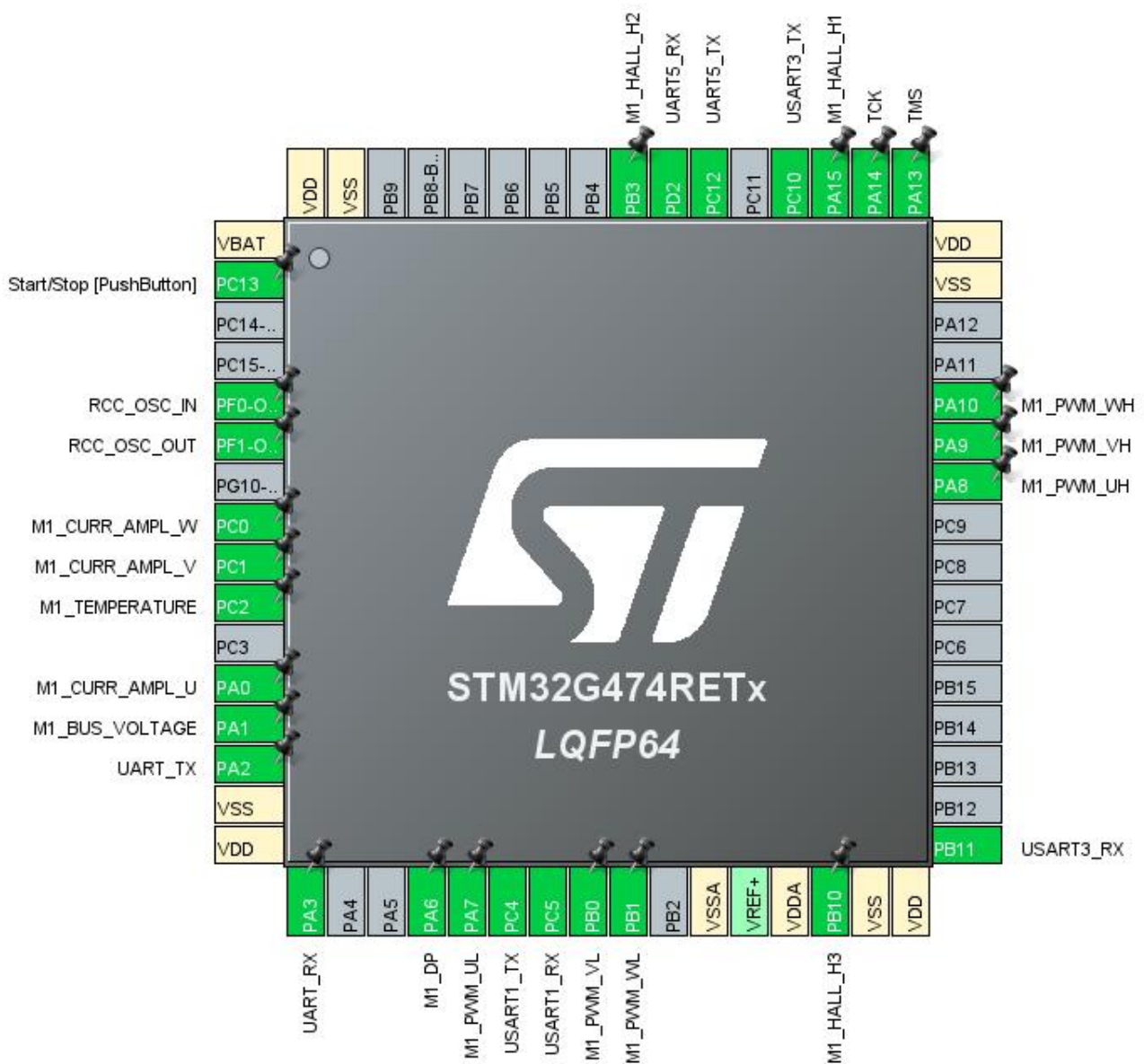
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration

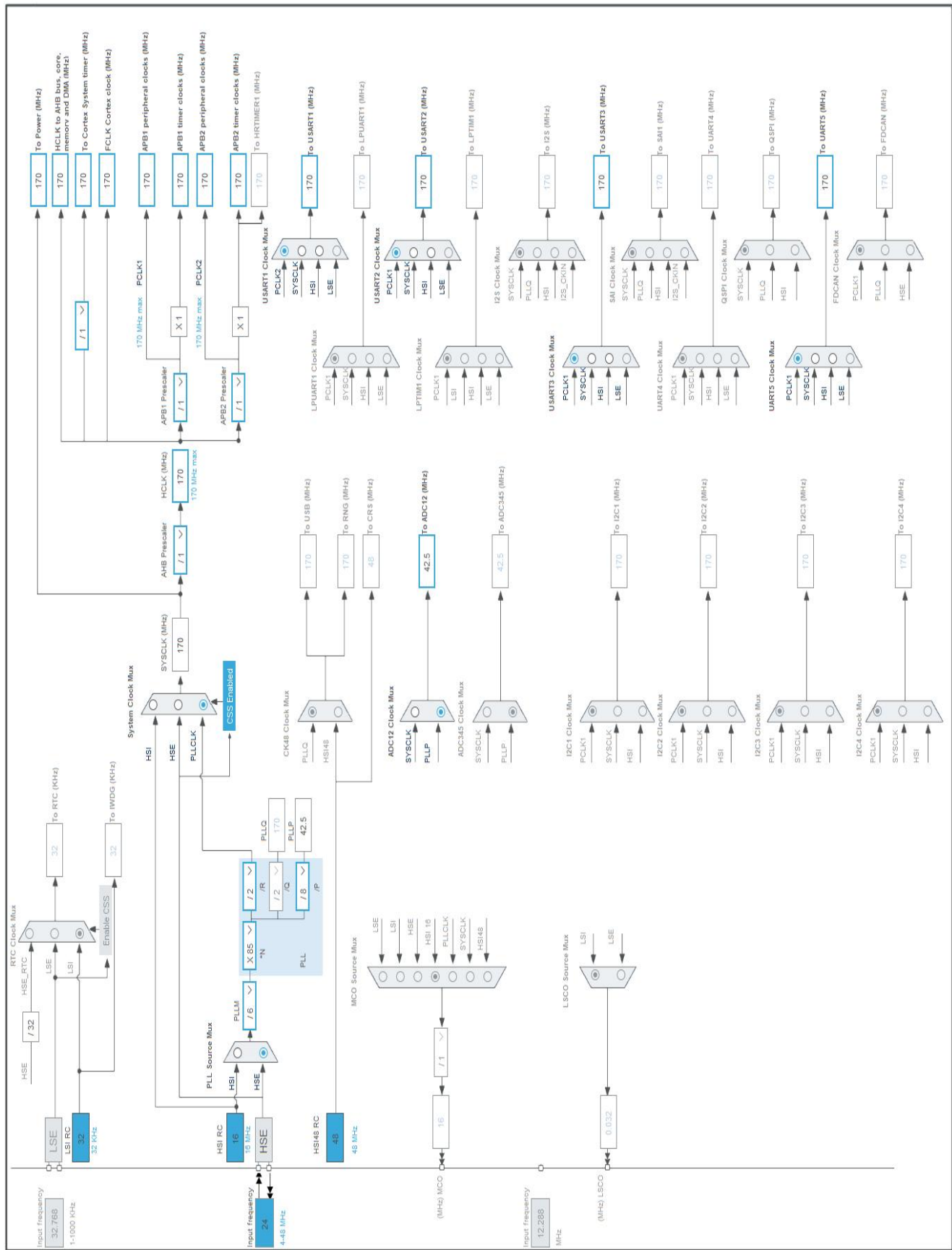


3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13	I/O	GPIO_EXTI13	Start/Stop [PushButton]
5	PF0-OSC_IN	I/O	RCC_OSC_IN	
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	
8	PC0	I/O	ADC2_IN6	M1_CURR_AMPL_W
9	PC1	I/O	ADC1_IN7, ADC2_IN7	M1_CURR_AMPL_V
10	PC2	I/O	ADC1_IN8	M1_TEMPERATURE
12	PA0	I/O	ADC1_IN1	M1_CURR_AMPL_U
13	PA1	I/O	ADC1_IN2	M1_BUS_VOLTAGE
14	PA2	I/O	USART2_TX	UART_TX
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	USART2_RX	UART_RX
20	PA6	I/O	TIM1_BKIN	M1_DP
21	PA7	I/O	TIM1_CH1N	M1_PWM_UL
22	PC4	I/O	USART1_TX	
23	PC5	I/O	USART1_RX	
24	PB0	I/O	TIM1_CH2N	M1_PWM_VL
25	PB1	I/O	TIM1_CH3N	M1_PWM_WL
27	VSSA	Power		
29	VDDA	Power		
30	PB10	I/O	TIM2_CH3	M1_HALL_H3
31	VSS	Power		
32	VDD	Power		
33	PB11	I/O	USART3_RX	
42	PA8	I/O	TIM1_CH1	M1_PWM_UH
43	PA9	I/O	TIM1_CH2	M1_PWM_VH
44	PA10	I/O	TIM1_CH3	M1_PWM_WH
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	TMS
50	PA14	I/O	SYS_JTCK-SWCLK	TCK
51	PA15	I/O	TIM2_CH1	M1_HALL_H1
52	PC10	I/O	USART3_TX	
54	PC12	I/O	UART5_TX	
55	PD2	I/O	UART5_RX	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
56	PB3	I/O	TIM2_CH2	M1_HALL_H2
63	VSS	Power		
64	VDD	Power		

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.6

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

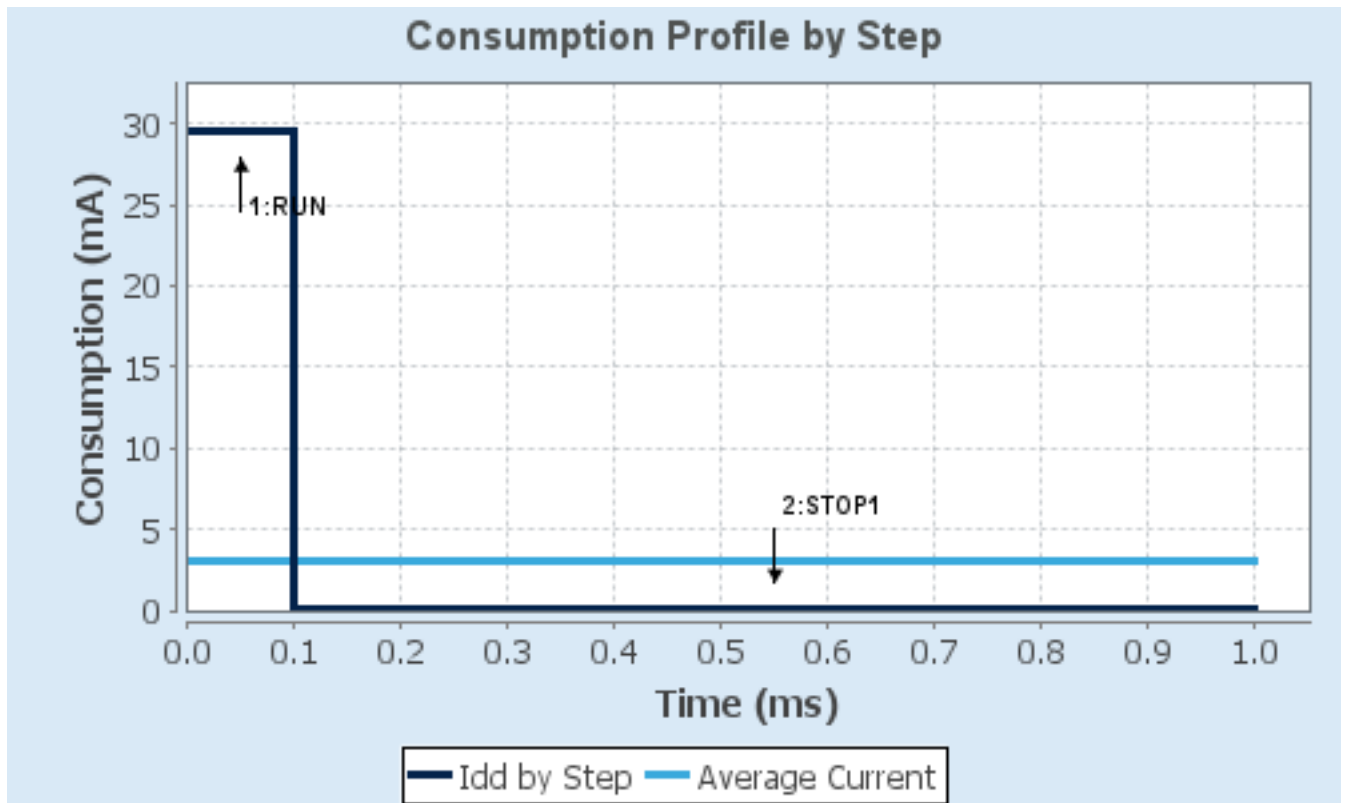
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.6	3.6
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	81.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	123.1	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	EIC_TuckRun_Hall_v6_4
Project Folder	D:\01d_MCU_G474\ST_TuckRunv6.4\EIC_TuckRun_Hall_v6_4
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.1
Application Structure	Basic
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC1_Init	ADC1
5	MX_ADC2_Init	ADC2
6	MX_CORDIC_Init	CORDIC
7	MX_TIM1_Init	TIM1
8	MX_TIM2_Init	TIM2
9	MX_USART2_UART_Init	USART2
10	MX_MotorControl_Init	MotorControl
11	MX_USART3_UART_Init	USART3

Rank	Function Name	Peripheral Instance Name
12	MX_UART5_Init	UART5
13	MX_USART1_UART_Init	USART1

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN7: IN7 Single-ended

IN8: IN8 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler **Asynchronous clock mode divided by 1 ***

Resolution ADC 12-bit resolution

Data Alignment **Left alignment ***

Gain Compensation 0

Scan Conversion Mode Enabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion **2 ***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel **Channel 2 ***

Sampling Time **47.5 Cycles ***

Offset Number No offset

Rank **2 ***

Channel **Channel 8 ***

Sampling Time **47.5 Cycles ***

Offset Number No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions **Enable ***

Number Of Conversions	2 *
External Trigger Source	Timer 1 Trigger Out event *
External Trigger Conversion Edge	Trigger detection on the rising edge
Enable Injected Oversampling	Disable
Injected Conversion Mode	None
Injected Queue	Injected Queue Disable
<u>Rank</u>	1
Channel	Channel 1
Sampling Time	6.5 Cycles *
Offset Number	No offset
<u>Rank</u>	2 *
Channel	Channel 7 *
Sampling Time	6.5 Cycles *
Offset Number	No offset
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. ADC2

IN6: IN6 Single-ended

IN7: IN7 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode
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ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1 *
Resolution	ADC 12-bit resolution
Data Alignment	Left alignment *
Gain Compensation	0
Scan Conversion Mode	Enabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled

DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data preserved
ADC_Regular_ConversionMode:	
Enable Regular Conversions	Disable *
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Enable *
Number Of Conversions	2 *
External Trigger Source	Timer 1 Trigger Out event *
External Trigger Conversion Edge	Trigger detection on the rising edge
Enable Injected Oversampling	Disable
Injected Conversion Mode	None
Injected Queue	Injected Queue Disable
<u>Rank</u>	1
Channel	Channel 7 *
Sampling Time	6.5 Cycles *
Offset Number	No offset
<u>Rank</u>	2 *
Channel	Channel 6
Sampling Time	6.5 Cycles *
Offset Number	No offset
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.3. CORDIC

mode: Activated

3.4. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.4.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled

Prefetch Buffer	Enabled *
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)
RCC Parameters:	
HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost
Peripherals Clock Configuration:	
Generate the peripherals clock configuration	TRUE

3.5. SYS

Debug: Serial Wire

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.6. TIM1

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Channel4: PWM Generation No Output

mode: Activate-Break-Input

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	$((TIM_CLOCK_DIVIDER) - 1) *$
Counter Mode	Center Aligned mode1 *
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	$((PWM_PERIOD_CYCLES) / 2) *$
Internal Clock Division (CKD)	Division by 2 *
Repetition Counter (RCR - 16 bits value)	(REP_COUNTER) *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Output Compare (OC4REF) *
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Enable
BRK Polarity	High
BRK Filter (4 bits value)	3 *
BRK Sources Configuration	
- Digital Input	Enable
Break_IO mode selection	Break IO is an Input
Digital Input Polarity	Polarity Low *
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	3 *
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Enable *
Off State Selection for Idle Mode (OSSI)	Enable *
Lock Configuration	Off
DeadTime Preload	Disable
Dead Time	((DEAD_TIME_COUNTS) / 2) *
Asymmetrical DeadTime	Disable
Falling Dead Time	0

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
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Pulse (16 bits value)	$((\text{PWM_PERIOD_CYCLES}) / 4) *$
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
Pulse (16 bits value)	$((\text{PWM_PERIOD_CYCLES}) / 4) *$
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	$((\text{PWM_PERIOD_CYCLES}) / 4) *$
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 2 *
Pulse (16 bits value)	$((\text{PWM_PERIOD_CYCLES}) / 2) - (\text{HTMIN})) *$
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.7. TIM2

Clock Source : Internal Clock

Combined Channels: XOR ON / Hall Sensor Mode

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	M1_HALL_TIM_PERIOD *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection TRGO	Output Compare (OC2REF)
------------------------------	-------------------------

Hall Sensor:

Prescaler Division Ratio	No division
Polarity	Rising Edge
Input Filter	M1_HALL_IC_FILTER *
Commutation Delay	0

3.8. UART5

Mode: Asynchronous

3.8.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable

MSB First

Disable

3.9. USART1

Mode: Asynchronous

3.9.1. Parameter Settings:

Basic Parameters:

Baud Rate	9600 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.10. USART2

Mode: Asynchronous

3.10.1. Parameter Settings:

Basic Parameters:

Baud Rate	1843200 *
Word Length	8 Bits (including Parity)
Parity	None

Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.11. USART3

Mode: Asynchronous

3.11.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable

Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.12. MotorControl

mode: Enabled

3.12.1. Parameter Settings:

M1_CURRENT_SENSING_TOPO	THREE_SHUNT *
M1_DTCOMP_SUPPORT	false
M2_DTCOMP_SUPPORT	false
M1_SPEED_SENSOR	HALL_SENSOR *
M2_SPEED_SENSOR	
Motor parameters:	
M1_MOTOR_ZEST_THRESHOLD_FREQ_HZ	0.0
M1_MOTOR_ZEST_INJECT_FREQ	0.0
M1_MOTOR_ZEST_INJECT_D	0.0
M1_MOTOR_ZEST_GAIN_D	0.0
M1_MOTOR_ZEST_GAIN_Q	0.0
M1_MOTOR_MASS_COPPER_KG	0.2
M1_MOTOR_COOLINGTAU_S	100.0
M1_MOTOR_INERTIA	0.0009
M1_MOTOR_FRICTION	0.01
M1_POLPULSES_PULSE_CURRENT_GOAL	0.0
M1_MOTOR_DYNAMICQGAIN	false

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC1	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	M1_CURR_AMPL_V
	PC2	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	M1_TEMPERATURE
	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	M1_CURR_AMPL_U
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	M1_BUS_VOLTAGE
ADC2	PC0	ADC2_IN6	Analog mode	No pull-up and no pull-down	n/a	M1_CURR_AMPL_W
	PC1	ADC2_IN7	Analog mode	No pull-up and no pull-down	n/a	M1_CURR_AMPL_V
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	TMS
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	TCK
TIM1	PA6	TIM1_BKIN	Alternate Function Open Drain	Pull-up *	Low	M1_DP
	PA7	TIM1_CH1N	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_UL
	PB0	TIM1_CH2N	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_VL
	PB1	TIM1_CH3N	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_WL
	PA8	TIM1_CH1	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_UH
	PA9	TIM1_CH2	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_VH
	PA10	TIM1_CH3	Alternate Function Push Pull	Pull-down *	High *	M1_PWM_WH
TIM2	PB10	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	High *	M1_HALL_H3
	PA15	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	High *	M1_HALL_H1
	PB3	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	High *	M1_HALL_H2
UART5	PC12	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD2	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PC4	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC5	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	UART_TX
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	UART_RX
USART3	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PC13	GPIO_EXTI13	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	Start/Stop [PushButton]

4.2. DMA configuration

DMA request	Stream	Direction	Priority
USART2_RX	DMA1_Channel1	Peripheral To Memory	Low
USART2_TX	DMA1_Channel2	Memory To Peripheral	Low
USART3_RX	DMA1_Channel3	Peripheral To Memory	Low
USART3_TX	DMA1_Channel4	Memory To Peripheral	Low
UART5_TX	DMA1_Channel5	Memory To Peripheral	Low
USART1_RX	DMA1_Channel6	Peripheral To Memory	Low

USART2_RX: DMA1_Channel1 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

USART2_TX: DMA1_Channel2 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

USART3_RX: DMA1_Channel3 DMA request Settings:

Mode: **Circular ***
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

USART3_TX: DMA1_Channel4 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***

Peripheral Data Width: Byte
Memory Data Width: Byte

UART5_TX: DMA1_Channel5 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART1_RX: DMA1_Channel6 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	4	0
DMA1 channel1 global interrupt	true	3	0
DMA1 channel3 global interrupt	true	3	0
DMA1 channel4 global interrupt	true	3	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel6 global interrupt	true	4	0
ADC1 and ADC2 global interrupt	true	2	0
TIM1 break interrupt and TIM15 global interrupt	true	4	1
TIM1 update interrupt and TIM16 global interrupt	true	0	0
TIM2 global interrupt	true	2	0
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	true	4	1
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	true	3	1
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28	true	3	1
EXTI line[15:10] interrupts	true	3	0
UART5 global interrupt / UART5 wake-up interrupt through EXTI line 35	true	5	1
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
DMA1 channel2 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM1 capture compare interrupt	unused		
FPU global interrupt	unused		
CORDIC interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	false	true
Hard fault interrupt	false	false	false
Memory management fault	false	false	false
Prefetch fault, memory access fault	false	false	false
Undefined instruction or illegal state	false	false	false
System service call via SWI instruction	false	false	false
Debug monitor	false	false	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 channel1 global interrupt	true	false	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
ADC1 and ADC2 global interrupt	true	false	true
TIM1 break interrupt and TIM15 global interrupt	true	false	true
TIM1 update interrupt and TIM16 global interrupt	true	false	true
TIM2 global interrupt	true	false	true
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	false	true	true
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	true	false	true
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28	false	true	true
EXTI line[15:10] interrupts	true	false	true
UART5 global interrupt / UART5 wake-up interrupt through EXTI line 35	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware							
MotorControl ✓							
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Utilities
DMA ✓	ADC1 ✓	TIM1 ✓	UART5 ✓			CORDIC ✓	
GPIO ✓	ADC2 ✓	TIM2 ✓	USART1 ✓				
IIVIC ✓			USART2 ✓				
RCC ✓			USART3 ✓				
SYS ✓							

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/fldpstpf11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-

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Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5738-stm32g4-series-lifetime-estimates-stmicroelectronics.pdf

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for related Tools obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-
& Software application-stmicroelectronics.pdf

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for related Tools inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
& Software

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for related Tools processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
& Software

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for related Tools guide-for-the-xcubesbsfu-stm32cube-expansion-package-
& Software stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5305-digital-filter-
for related Tools implementation-with-the-fmac-using-stm32cubeg4-mcu-package-
& Software stmicroelectronics.pdf

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for related Tools firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
& Software

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for related Tools rgb-led-control-using-the-bg474edpow1-discovery-kit-
& Software stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5360-getting-started-
for related Tools with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-
& Software stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5361-getting-started-
for related Tools with-projects-based-on-dualcore-stm32h7-microcontrollers-in-
& Software stm32cubeide-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5394-getting-started-
for related Tools with-projects-based-on-the-stm32l5-series-in-stm32cubeide-
& Software stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5418-how-to-build-a-
for related Tools simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
& Software

Application Notes https://www.st.com/resource/en/application_note/an5426-migrating-
for related Tools graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-

& Software	550-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dual-core-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5785-boost-voltage-mode-on-bg474edpow1-discovery-kit-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smmps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4502-stm32-smbus-pmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5496-guidelines-for-the-buck-voltage-mode-on-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf
& Software	

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5497-introduction-to-the-buck-current-mode-with-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00431551.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
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Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf

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Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
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